

A drone that finds people from one sentence

SAGE-UAV

Semantic AI-Guided Exploration & Active Search

An autonomous drone that understands "find all people in this area", searches, detects, localizes in 3D, verifies, reports and lands by itself.

89%

of people found

0.46 m

mean position error

89%

precision

10 min

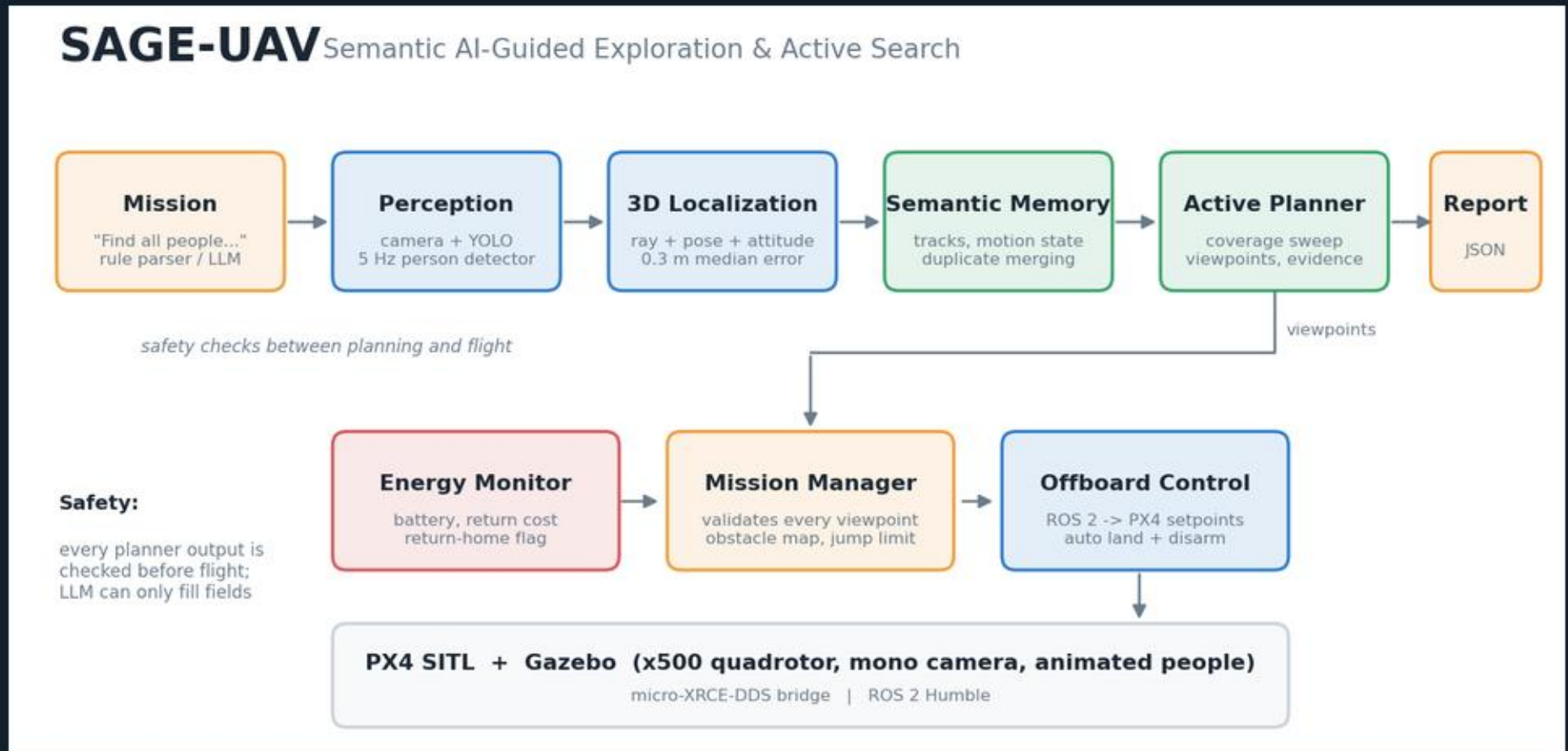
to sweep 24 x 24 m

ROS 2 | PX4 | Gazebo | YOLO | semantic mapping | active perception

Simulation results, 7 scored runs (see docs/limitations.md)

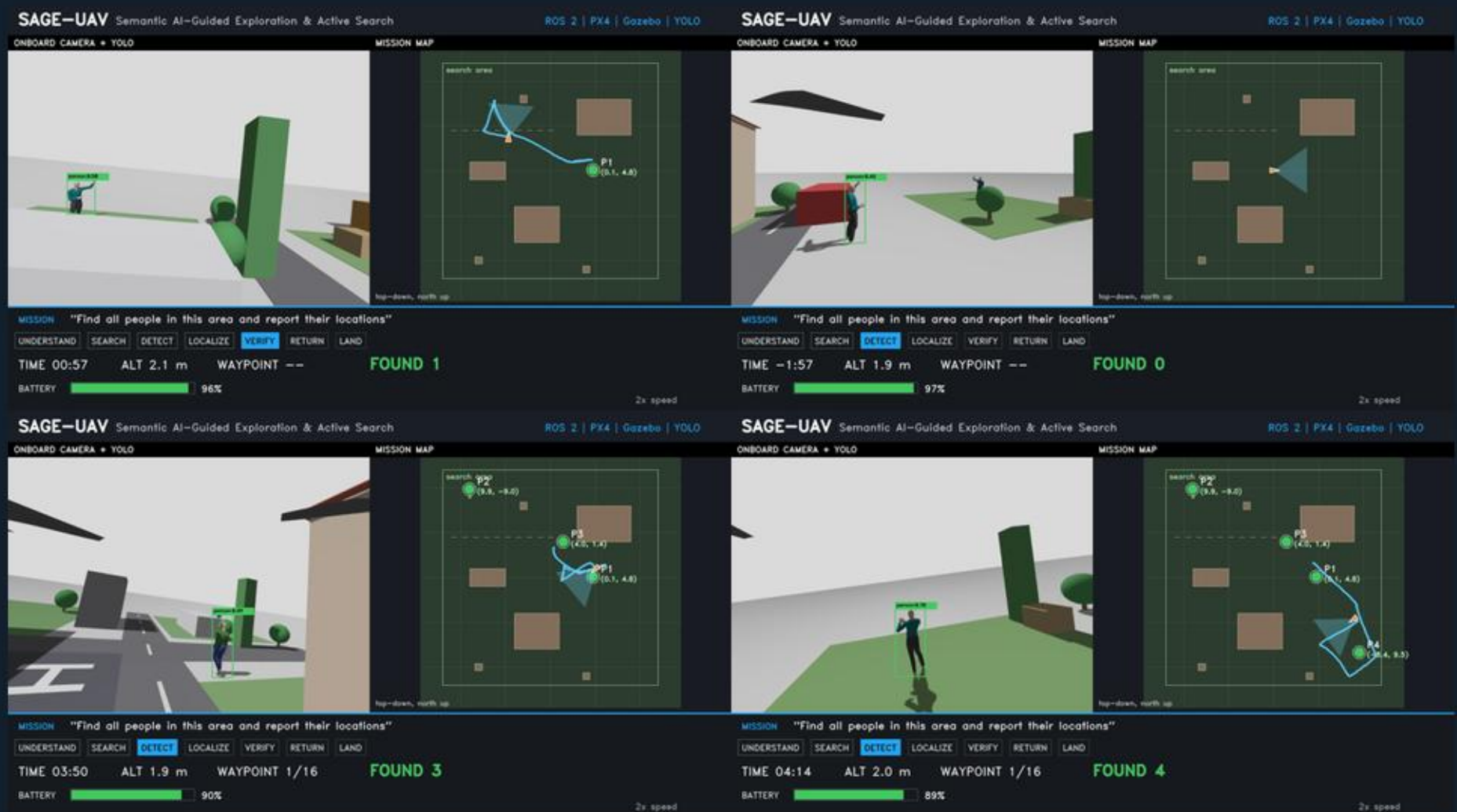
"Find all people in this area and report their locations": it searches, detects, locates people in 3D, verifies, and lands itself.

How it works



Mission text becomes a validated spec. Perception, 3D localization and a semantic memory feed an active planner; every viewpoint passes an independent safety check before it reaches PX4.

The flight



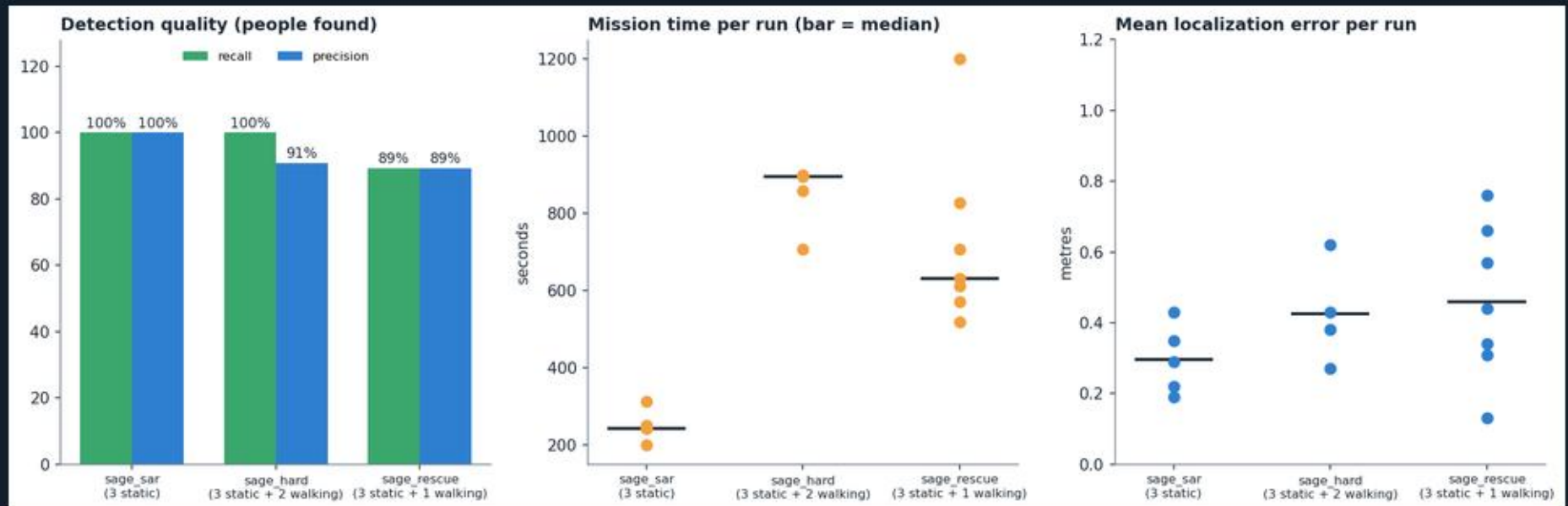
Onboard camera with YOLO detections and the live mission map: search, first detection, and people verified one by one.

Seen from above the drone



A Gazebo camera mounted above the drone shows the real drone model (north-up, with a mini map). Same flight, 2x speed.

Results in simulation



Scored automatically against ground truth: people found, precision, position error, and mission time over repeated runs.

Open source

github.com/haidar996/SAGE-UAV

ROS 2 | PX4 | Gazebo | YOLO
semantic mapping | active perception

Code, worlds, results, setup guide and the demo videos are in the repository (MIT).